

CADABRA AI CAD Copilot

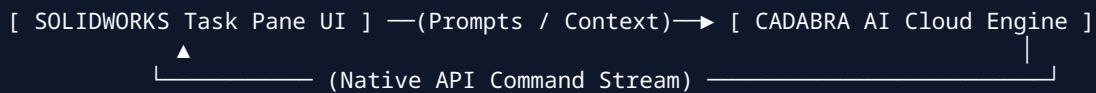
Executive Technical Specification, Architecture & Market Comparison Guide

01. WHAT IS CADABRA?

CADABRA is an AI-powered CAD Copilot built natively as an add-in for **SOLIDWORKS**. Unlike generic mesh or generative visualizers, CADABRA interprets natural language prompt engineering and generates fully parametric, native CAD features directly inside the SOLIDWORKS graphics tree.

It acts as a real-time engineering partner that writes fully editable sketch dimensions, extrusions, revolve cuts, hole wizard placements, mate constraints, and automated 2D engineering drawings.

02. TECHNICAL ARCHITECTURE & HOW IT WORKS



Frontend Functionality (SOLIDWORKS UI)

- **Task Pane Integration:** Native side panel operating inside SOLIDWORKS with zero context-switching.
- **Multi-Operating Modes:**
 - *Ask Mode:* Conceptual DFM advice, design standard verification.
 - *Agent Mode:* Parametric CAD generation and feature tree building.
 - *Debug Mode:* Auto-fixes broken rebuild errors & mates.
 - *Drawings Mode:* Generates 2D sheets, BOMs & GD&T.

Backend Architecture (Cloud Engine)

- **Contextual CAD Parsing:** Reads current CAD state, planes, mates, units, and references securely.
- **Reasoning & API Mapping:** High-reasoning AI converts human intent into precise SOLIDWORKS API / C# commands.
- **Guarded Execution Loop:** Executes operations in rounds. Catches rebuild failures, rolls back, auto-corrects parameters, and re-executes.

03. OUTPUT & SYSTEM CAPABILITIES

- **Parametric Native 3D Models:** Creates standard, editable sketches, features, extrusions, revolves, fillets, and patterns.
- **Assembly & Mate Intelligence:** Positions components, applies mate constraints, and resolves interferences automatically.
- **2D Drawings & BOM Generation:** Produces production-ready 2D drawing sheets with title blocks, GD&T, and Bill of Materials.
- **DFM & Manufacturing Cost Estimation:** Analyzes parts for CNC machining/3D printing constraints and estimates pricing.
- **Multi-Format File Exporting:** Direct export capabilities to STEP, IGES, Parasolid, STL, 3MF, and PDF formats.

04. PRODUCT COMPARISON (CADABRA VS. ALTERNATIVES)

Feature / Dimension	CADABRA AI	3D Mesh Generators (Suma/ Meshy)	Generative Design (Fusion/ Creo)
Output Format	Native, editable SOLIDWORKS Feature Tree	Uneditable 3D Mesh Blobs (.STL / .OBJ)	FEA-optimized organic mesh shapes
Integration Location	Native SOLIDWORKS Add-In Task Pane	External Web Application / Browser	Built-in Cloud Computation Module
Primary Input	Natural language English prompts	Text Prompts / Concept Images	Physics load cases & constraints
2D Drawings & BOM	Fully automated with company templates	Not supported	Manual conversion required
Target Application	Mechanical engineering & CAD automation	Gaming assets & visual concepts	Structural weight reduction

05. PRICING TIERS & MARKET ALTERNATIVES

CADABRA Pricing Tiers

- **Free Plan** **\$0/mo** — 700K credits/month (Ask Mode & search).
- **Student** **~\$15/mo** — 3M credits/month (Agent Mode & Text-to-CAD).
- **Standard** **\$50/mo** (\$500/yr) — 5M credits/month (Commercial usage).
- **Pro** **\$99-\$120/mo** — 10M credits/month (DFM & Cost estimation).
- **Enterprise** **Custom Quote** — Unlimited credits, custom templates & SSO.

Market Available Alternatives

- **MorphoAI:** AI assistant focusing on parametric CAD feature generation.
- **Aurorin CAD:** Mechanical design AI copilot for sketching acceleration.
- **MecAgent:** AI mechanical assistant for standard part selection & assemblies.
- **Leo AI / Zoo Text-to-CAD:** Open API converting prompts to STEP / KCL formats.